Example 7.13

A pair of dice is thrown 4 times. If getting a doublet is considered a success, find the probability of 2 successes.

Solution :

In a throw of a pair of dice the doublets are (1,1) (2,2) (3,3) (4,4) (5,5) (6,6)

Probability of getting a doublet

$p = 6/36 = 1/6$

$\Rightarrow q = 1 - p = 5/6$ and also $n = 4$ is given

The probabitliy of successes

$$= \binom{4}{x}\left(\frac{1}{6}\right)^{x}\left(\frac{5}{6}\right)^{4-x}$$

Therefore the probability of 2 successes are

$$P(X = 2) = \binom{4}{2}\left(\frac{1}{6}\right)^{2}\left(\frac{5}{6}\right)^{4-2}$$

$$= 6 \times \frac{1}{36} \times \frac{25}{36}$$

$$= \frac{25}{216}$$

Exercise 7.1

1. Define Binomial distribution.
2. Define Bernoulli trials.
3. Derive the mean and variance of binomial distribution.
4. Write down the conditions for which the binomial distribution can be used.
5. Mention the properties of binomial distribution.
6. If 5% of the items produced turn out to be defective, then find out the probability that out of 20 items selected at random there are

   (i) exactly three defectives

   (ii) atleast two defectives

   (iii) exactly 4 defectives

   (iv) find the mean and variance
7. In a particular university 40% of the students are having news paper reading habit. Nine university students are selected to find their views on reading habit. Find the probability that

   (i) none of those selected have news paper reading habit

   (ii) all those selected have news paper reading habit

   (iii) atleast two third have news paper reading habit.
8. In a family of 3 children, what is the probability that there will be exactly 2 girls?
9. Defects in yarn manufactured by a local mill can be approximated by a distribution with a mean of 1.2 defects for every 6 metres of length. If lengths of 6 metres are to be inspected, find the probability of less than 2 defects.
10. If 18% of the bolts produced by a machine are defective, determine the probability that out of the 4 bolts chosen at random

    (i) exactly one will be defective

    (ii) none will be defective

    (iii) atmost 2 will be defective
11. If the probability of success is 0.09, how many trials are needed to have a probability of atleast one success as 1/3 or more ?
12. Among 28 professors of a certain department, 18 drive foreign cars and 10 drive local made cars. If 5 of these professors are selected at random, what is the probability that atleast 3 of them drive foreign cars?

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